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Cleaning wheel including reels and transmission for cleaning film

Abstract

A cleaning equipment includes a first take-up reel, a first supply reel, a frame, a first cleaning film, and a transmission assembly. The frame includes an inner space accommodating the first take-up reel and the first supply reel side by side and an outer surface having at least one gap communicating with the inner space. The first cleaning film has a supply part on the first supply reel, a take-up part on the first take-up reel, and a middle part at least partially covering the outer surface of the frame. Two ends of the middle part enter the inner space through the at least one gap and are connected to the supply part and the take-up part, respectively. The transmission assembly is coupled to the first supply reel and the first take-up reel and drives at least one of the first supply reel and the first take-up reel to rotate.

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Background/Summary

CROSS REFERENCE TO THE RELATED APPLICATION (1) The application claims the benefit of U.S. Provisional Application No. 63/292,600, filed on Dec. 22, 2021 and Taiwan Application No. 111126172, filed on Jul. 12, 2022. The entirety of each of the mentioned above patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a cleaning equipment and a cleaning method. Specifically, the present invention relates to a non-blow/suction type dust collecting cleaning equipment with an inner roll structure and a cleaning method thereof.

2. Description of the Prior Art

(2) With the continuous advancement of science and technology, all kinds of machinery and equipment are also moving towards the goal of complete automation. To achieve the purpose of reducing manpower requirements, autonomous cleaning equipment is gradually introduced in both home and factory cleaning. For example, the current common sweeping robots have become the home cleaning choice for many users. At present, the common automatic floor cleaning equipment on the market are all suction/blow type dust collectors, that is, airflow is generated through pressure to inhale or blow away dust and garbage. However, the airflow generated by suction/blow dust collectors may cause turbulence in areas that require strict control of the indoor environment (such as clean rooms, laboratories, etc.), thereby causing secondary pollution.

(3) In addition to the suction/blow type cleaning equipment, there are also adhesive cleaning rollers on the market. These cleaning rollers mainly rely on the adsorptive capacity of the surface of the adhesive paper to remove the dirt without generating airflow like the suction/blow type cleaning equipment. However, the use of such products usually requires manual operation, and when the surface of the cleaning roller no longer has the adsorptive capacity, the cleaning paper on the surface needs to be manually removed for continued use. Therefore, although the current adhesive roller can avoid the problem of turbulence, it cannot achieve the purpose of automation to reduce manpower requirements. Therefore, cleaning equipment and method that is automated and does not cause turbulence during cleaning would be desirable in the industry.

SUMMARY OF THE INVENTION

(4) The present invention is to provide cleaning equipment and method, which can achieve automatic target surface (e.g., ground) cleaning.

(5) The present invention is to provide cleaning equipment and method, which can avoid causing turbulent flow, thereby preventing secondary pollution to the equipment in the clean room, laboratory, and other spaces.

(6) In one embodiment, the cleaning equipment includes a first take-up reel; a first supply reel; a frame, comprising an outer surface and an inner space located inside the outer surface, wherein the outer surface has at least one gap communicating with the inner space, and the inner space accommodates the first take-up reel and the first supply reel side by side; a first cleaning film with a supply part, a take-up part and a middle part, wherein the first cleaning film has adsorptive capacity, the supply part is arranged on the first supply reel, the take-up part is arranged on the first take-up reel, at least part of the middle part covers the outer surface of the frame, and two ends of the middle part respectively enter the inner space through the at least one gap and are respectively

connected to the supply part and the take-up part; and a transmission assembly coupled to the first supply reel and the first take-up reel and configured to drive at least one of the first supply reel and the first take-up reel to rotate.

(7) In one embodiment, the cleaning method includes pulling at least part of a first cleaning film arranged on a first supply reel out to leave an inner space of a frame through at least one gap of the frame, cover an outer surface of the frame, and then enter the inner space again through the at least one gap and be arranged on a first take-up reel, wherein the part of the first cleaning film arranged on the first supply reel becomes a supply part, the part covering the outer surface becomes a middle part, and the part arranged on the first take-up reel becomes a take-up part, the first supply reel and the first take-up reel are arranged side by side in the inner space, and the first cleaning film has absorptive capacity; and operating a transmission assembly coupled to the first supply reel and the first take-up reel to drive at least one of the first supply reel and the first take-up reel to rotate, wherein in response to rotation of the first take-up reel, at least part of the middle part becomes the take-up part, and in response to rotation of the first supply reel, at least part of the supply part leaves the first supply reel and becomes at least part of the middle part.

Description

BRIEF DESCRIPTION OF THE APPENDED DRAWINGS

- (1) FIG. 1A and FIG. 1B show schematic diagrams of different angles of the cleaning equipment according to a first embodiment of the present invention.
- (2) FIG. 2 shows a schematic diagram of the cleaning equipment with the second take-up reel and the second supply reel according to a second embodiment of the present invention.
- (3) FIG. 3A and FIG. 3B show schematic diagrams of the cleaning equipment with the cleaning wheel according to a third embodiment of the present invention.
- (4) FIG. 4A to FIG. 4C show schematic diagrams illustrating a cleaning process performed by the cleaning equipment according to the third embodiment of the present invention.
- (5) FIG. 5A shows a schematic exploded diagram of the cleaning equipment according to the third embodiment of the present invention.
- (6) FIG. 5B shows a schematic exploded diagram of the transmission assembly shown in FIG. 5A.
- (7) FIG. 6A shows a schematic diagram of the interior of the gear plate of the cleaning equipment according to the third embodiment of the present invention.
- (8) FIG. 6B shows a rear side view of the gear plate of the cleaning equipment according to the third embodiment of the present invention.
- (9) FIGS. 7A to FIG. 7C show schematic diagrams of fixing the limit element according to the third embodiment of the present invention.
- (10) FIG. 8A and FIG. 8B show schematic diagrams of operation mode 1 of the transmission assembly according to the third embodiment of the present invention.
- (11) FIG. 8C and FIG. 8D show schematic diagrams of operation mode 2 of the transmission assembly according to a fourth embodiment of the present invention.
- (12) FIG. 9A shows a schematic diagram of the lifting module according to the third embodiment of the present invention.
- (13) FIG. 9B shows a schematic diagram of the lifting process performed by the lifting module according to the third embodiment of the present invention.
- (14) FIG. 10A to FIG. 10D show a schematic diagram of the replacement of the cleaning film covering the outer surface according to the third embodiment of the present invention.
- (15) FIG. 11A shows a schematic diagram of the procedure for determining the adsorptive capacity of the cleaning film according to the third embodiment of the present invention.
- (16) FIG. 11B and FIG. 11C show schematic diagrams of images captured and processed by the

monitoring unit according to the third embodiment of the present invention.

(17) FIG. 12 shows a schematic diagram of the anti-reverse element according to the third embodiment of the present invention.

(18) FIG. 13 shows a flow chart of the cleaning method according to a fifth embodiment of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

(19) The present invention is described below through specific embodiments and the accompanying drawings, and those skilled in the art can understand the advantages and effects of the present invention from the contents disclosed in this specification. However, the content disclosed below is not intended to limit the scope of the present invention, and those skilled in the art can implement the present invention with other different embodiments without departing from the spirit of the present invention.

(20) In the drawings, the sizes of various components and modules are enlarged for clarity. Throughout the specification, the same reference numerals refer to the same elements. It will be understood that when an element is referred to as being “on” or “connected to” another element, it can be directly on or connected to the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or “directly connected to” another element, there are no intervening elements present. As used herein, “connected” may refer to a physical and/or electrical connection. Furthermore, “electrically connected” or “coupled” may refer to the existence of other elements between the two elements.

(21) It will be understood that, although the terms “first”, “second”, and “third” may be used herein to describe various elements, components, regions, layers, and/or parts, these elements, components, regions, and/or parts shall not be limited by these terms. These terms are only used to distinguish one element, component, region, layer, or part from another element, component, region, layer, or part. Thus, “first supply reel”, “first take-up reel” or “first cleaning film” discussed below may be referred to as a second supply reel, take-up reel, or cleaning film without departing from the teachings herein.

(22) The term used herein is to describe particular embodiments only and is not limiting. As used herein, the singular forms “a”, “an”, and “the” are intended to include the plural forms, including “at least one” unless the content dictates otherwise. “Or” means “and/or”. As used herein, the term “and/or” includes any combinations of one or more of the associated listed items. It will also be understood that, when used in this specification, the terms “comprising” and/or “including” designate the presence or addition of stated features, regions, integers, steps, operations, elements, and/or components, but do not exclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components and/or combinations thereof.

(23) Referring to FIG. 1A, the cleaning equipment **100** according to the first embodiment of the present invention is illustrated. The cleaning equipment **100** includes a hollow cylindrical frame **110**. Frame **110** includes an outer surface **111**, an inner space I, and a gap **113**. The inner space I is the hollow part of the cylindrical frame **110**, and the gap **113** extends along the axial direction of the cylindrical frame **110** and communicates with the inner space I. The inner space I accommodates the first take-up reel **130** and the first supply reel **150**. The first take-up reel **130** is arranged with the take-up part **173** of the first cleaning film **170**, and the first supply reel **150** is arranged with the supply part **175** of the first cleaning film **170**. In this embodiment, “arranged” may refer to winding the take-up part **173** and the supplying part **175** respectively round the axes of the first take-up reel **130** by clockwise and the first supply reel **150** counterclockwise. At least part of the middle part **171** of the first cleaning film **170** covers the outer surface **111** and enters the inner space I through the gap **113** to connect the supply part **175** and the take-up part **173**, respectively. The first cleaning film **170** has adsorptive capacity. It should be noted that the adsorptive capacity described herein may be the adsorptive capacity caused by contact or the adsorptive capacity not caused by contact (e.g., electrostatic force), which is not limited in the

present invention.

(24) Continuing to refer to the variant embodiment shown in FIG. 1B, the cleaning equipment **100** further includes a transmission assembly **190**, which drives at least one of the first supply reel **150** and the first take-up reel **130** to rotate when it operates. In this embodiment, the transmission assembly **190** includes a belt **191** and a handle **193**. The belt **191** is simultaneously coupled to the first take-up reel **130** and the first supply reel **150** (e.g., coupled by the gears or gear plate at the ends of the first take-up reel **130** and the first supply reel **150**), and the handle **193** is coupled to the first supply reel **150**. In different embodiments, the handle **193** can be coupled to the first take-up reel **150**, which is not limited in the present invention. The operation of the transmission assembly **190** refers to that when the handle **193** is turned clockwise by an external force, the belt **191** drives the first take-up reel **130** to rotate clockwise in response to the clockwise rotation of the first supply reel **150**. Since the take-up part **173** and the supply part **175** wind round the first take-up reel **130** and the first supply reel **150** in a clockwise and counterclockwise manner respectively, when the first take-up reel **130** and the first supply reel **150** rotate clockwise at the same time, at least part of the supply part **175** leaves the first supply reel **150** to become the middle part **171**, and at least part of the middle part **171** is wound around the first take-up reel **130** to become the take-up part **173**. With this configuration, the middle part **171** originally covering the outer surface **111** is wound up by the first take-up reel **130**, and part of the supply part **173** arranged on the first supply reel **150** replaces the original middle part **171** and covers the outer surface **111**, so as to achieve the purpose of replacing the middle part of the first cleaning film **170** covering the outer surface **111**, i.e., alternately covering the outer surface **111** with the unwound portion of the first cleaning film **170**.

(25) It should be noted that the direction of the winding and rotation (i.e., clockwise or counterclockwise) are all exemplary, and other directions may be used in different embodiments. For example, the take-up part and the supply part can be respectively wound round counterclockwise and clockwise, and the first take-up reel **130** and the first supply reel **150** can rotate counterclockwise, which is not limited in the present invention.

(26) In addition, the cleaning equipment **100** may further include a guard plate **195** and an anti-stick strip **197**, wherein the guard plate **195** is disposed of at the end of the axial direction (one side or both sides) of the frame **110** to prevent dust and dirt from entering the inner space I. The anti-stick strip **197** is located in gap **113** and extends parallel to the extending direction (i.e., the axial direction) of frame **110** and gap **113**. Preferably, the anti-stick strip **197** does not protrude from the outer surface **111** and does not contact frame **110**. That is to say, the anti-stick strip **197** divides gap **113** into two parallel long gaps. In this embodiment, the anti-stick strip **197** is coupled to the guard plate **195** and is cylindrical, but in different embodiments, the anti-stick strip **197** can have different shapes (e.g., a diamond-shaped column), which is not limited in the present invention. With this configuration, both ends of the middle part **171** will be separated by the anti-stick strips **197** when they respectively enter the inner space I through the gap **113**, thereby preventing them from sticking to each other.

(27) Referring to FIG. 2, the cleaning equipment **200** according to the second embodiment of the present invention is illustrated. The cleaning equipment **200** has a structure similar to that of the cleaning equipment **100** of the first embodiment, and further includes a second take-up reel **230**, a second supply reel **250**, and a second cleaning film **270**. In addition, frame **110** includes a first gap **213a** and a second gap **213b**, and the first gap **213a** and the second gap **213b** divide the outer surface **111** into a first outer surface **211a** and a second outer surface **211b**. The second cleaning film **270** has the same structure as the first cleaning film **170** with a middle part **271**, a take-up part **273**, and a supply part **275**. The take-up part **273** is arranged on the second take-up reel **230**, and the supply part **275** is arranged on the second supply reel **250**. The middle parts **171** and **271** of the first cleaning film **170** and the second cleaning film **270** respectively cover the first outer surface **211a** and the second outer surface **211b**, and both ends of the middle parts **171** and **271** enter the

inner space I through the first gap **213a** and the second gap **213b** respectively, to connect the take-up parts **173**, **273** and the supply parts **175**, **275**. In addition, when the transmission assembly **190** operates, it can drive the first take-up reel **130**, the second take-up reel **230**, the first supply reel **150**, and the second supply reel **250** to rotate clockwise or counterclockwise (for details, refer to FIG. **8A** to FIG. **8D**), thereby achieving the purpose of alternatingly covering the outer surface **111** with unwound portions of the cleaning films (i.e., the first cleaning film **170** and the second cleaning film **270**) as described above.

(28) FIG. **3A** and FIG. **3B** illustrate a cleaning equipment **300** according to the third embodiment of the present invention. The cleaning equipment **300** may have a structure similar to that of the cleaning equipment **100** or **200**, and further includes a cleaning wheel **310**. It should be noted that, in order to concisely illustrate the present invention, the cleaning equipment **300** is shown in a manner similar to the cleaning equipment **100** (without the second take-up reel **230** and the second supply reel **250**). However, the third embodiment described below with the drawings can have the structure of the cleaning equipment **100** or **200**, which is not limited in the present invention.

(29) First, referring to FIG. **3A**, the cleaning wheel **310** is disposed under the frame **110** and contacts the middle part **171**, which covers the outer surface **111**. The adsorptive capacity of the cleaning wheel **310** is smaller than that of the first cleaning film **170**. Since the axial direction of the cleaning wheel **310** is parallel to the axial direction of the frame **110**, the entire cleaning equipment **300** is driven to move forward in a direction vertical to the axial direction of the frame **110** when the cleaning wheel **310** rotates, and the frame **110** is affected by the frictional force and rotates accordingly. It should be noted that although FIG. **3A** shows only one cleaning wheel **310**, in various embodiments, there may be a plurality of cleaning wheels **310** (e.g., referring to FIG. **3B** showing cleaning equipment **300** having two cleaning wheels **310**), which is not limited in the present invention.

(30) FIG. **4A** to FIG. **4C** continue illustrating the cleaning equipment **300** according to the third embodiment of the present invention. Since the cleaning wheel **310** has an adsorptive capacity, when the cleaning wheel **310** contacts the target surface (such as the ground) during the rotation, it will adsorb the dust, garbage, and the like. Then, since the adsorptive capacity of the cleaning wheel **310** is smaller than that of the first cleaning film **170**, when the part of the cleaning wheel **310** adsorbing the dirt contacts the middle part **171**, the dirt will be passed on to the first cleaning film **170**. With this configuration, the cleaning equipment **300** can adsorb the dirt on the target surface through the cleaning wheel **310**, and then pass on the dirt to the first cleaning film **170**, so that the cleaning wheel **310** maintains the adsorptive capacity, thereby achieving the purpose of cleaning the target surface.

(31) FIG. **5A** is a schematic exploded diagram of the detail between the frame **110** and the transmission assembly **190** according to an embodiment, and FIG. **5B** is a schematic exploded diagram of the transmission assembly **190** according to the embodiment shown in FIG. **5A**. In this embodiment, the transmission assembly **190** includes a gear plate **510** and a top plate **570**, wherein the gear plate **510** has teeth protruding toward the radial direction around it, and the middle of the surface paralleling to the radial direction is recessed to accommodate the take-up gear **513** and the supply gear **515** (refer to FIG. **5B**). The surfaces of the take-up gear **513** and the supply gear **515** both have a first tenon/mortise element **530**, which are disposed on the side not contact the gear plate (i.e., the side facing the frame **110**), and the first take-up reel **130** and the first supply reel **150** both have a second tenon/mortise element **550**, which is arranged at the end of the axial direction (i.e., the surface facing the gear plate **510**). The first tenon/mortise element **530** and the second tenon/mortise element **550** correspond to each other. Specifically, in this embodiment, the first tenon/mortise element **530** and the second tenon/mortise element **550** are respectively a polygonal convex piece and a polygonal concave piece, and the shapes and sizes are matched with each other. When the transmission assembly **190** is coupled to the frame **110**, the first tenon/mortise element **530** and the second tenon/mortise element **550** can be combined. With this configuration, when the

take-up gear **513** and the supply gear **515** rotate, the first tenon/mortise element **530** drives the second tenon/mortise element **550** to rotate, thereby causing the first take-up reel **130** and the first supply reel **150** to rotate, and to achieve the purpose of alternately covering the outer surface **111** with the unwound portion of the first cleaning film **170** as described above.

(32) On the other hand, the second take-up reel **230** and the second supply reel **250** may have the similar structure as the first take-up reel **130** and the first supply reel **150**. When the take-up gear **513** and the supply gear **515** rotate, the first tenon/mortise element **530** drives the second tenon/mortise element **550** to rotate, thereby causing the second take-up reel **230** and the second supply reel **250** to rotate accordingly, so as to achieve the purpose of alternately covering the outer surface **111** with the unwound portion of the second cleaning film **270**. It should be noted that, although the first tenon/mortise element **530** is shown as a convex piece and the second tenon/mortise element **550** is shown as a concave piece, in different embodiments, the first tenon/mortise element **530** and the second tenon/mortise elements **550** may be other mutually matched structures (e.g., the first tenon/mortise element **530** is a concave piece, and the second tenon/mortise element **550** is a convex piece), which is not limited in the present invention.

(33) Referring to FIG. 5B, the top plate **570** of the cleaning equipment **300** is coupled to the gear plate **510** to prevent dirt from entering the gear plate **510**. The top plate **570** has top-plate holes **571** (the total number of which corresponds to the first take-up reel **130**, the first supply reel **150**, the second take-up reel **230**, and the second supply reel **250**) located corresponding to the take-up gear **513** and the supply gear **515**, so that the first tenon/mortise element **530** can pass through the top-plate hole **571** and couple with the second tenon/mortise element **550**. It should be noted that although the top-plate hole **571** is shown in a circular shape in this embodiment, its purpose is only to allow the first tenon/mortise element **530** to pass through without being blocked by the top plate **570**, so it can be any suitable shape (for example, the same polygon as the first tenon/mortise element **530**), which is not limited in the present invention.

(34) Referring to FIG. 5A, the top plate **570** may further include a tenon **573** and an alignment parent element **575**, and in this embodiment, the cleaning equipment **300** may include a guard plate **195** disposed on both ends of the frame **110** as in the first embodiment. The guard plate **195** has holes to expose the ends of the first take-up reel **130** and the first supply reel **150**, and the guard plate **195** further includes a mortise **577** and an alignment child element **579** (see FIG. 5A). The tenon **573** corresponds to the mortise **577**, so that the transmission assembly **195** can be combined with the frame **110**, and the alignment parent element **575** corresponds to the alignment child element **579** (for example, the alignment parent element **575** can be a long strip protrusion, an alignment child element **579** can be a long strip groove of the same shape as the alignment parent element **575**) to facilitate quick alignment when combining the transmission assembly **190** with the frame **110**. It should be noted that the aforementioned arrangement of the tenon **573**, the mortise **577**, the alignment parent element **575**, and the alignment child element **579** are only exemplary, and in different embodiments, there may be other arrangements (for example, the tenon **573** is arranged in the guard plate **195**, and the mortise **577** is arranged on the top plate **570**, etc.), which is not limited in the present invention.

(35) FIG. 6A shows the internal structure of the gear plate **510** according to an embodiment, and FIG. 6B is a rear side view of the gear plate **510** of the embodiment shown in FIG. 6A. In this embodiment, the cleaning equipment **300** further includes a control element **610**. The control element **610** passes through the gear plate **510** and is provided with a control gear **611** inside the gear plate **510** and a limit element **613** outside the gear plate **510** (i.e., the side not provided with the take-up gear **513** and the supply gear **515**). For the description of the limit element, please refer to FIG. 7A, FIG. 7B, and FIG. 7C. The limit element **613** has a linkage with the control gear **611** (that is, when the limit element **613** is fixed, the control gear **611** will not be able to rotate; when the limit element **613** is rotated, the control gear **611** will rotate accordingly), and the control gear **611** meshes with the take-up gear **513** and the supply gear **515**, respectively.

(36) Referring to FIG. 7A the illustration of the cleaning equipment **300** according to the third embodiment of the present invention is continued. In this embodiment, the cleaning equipment **300** further includes a driving module **650**, which includes a fixer **655** for fixing the limit element **613**. FIG. 7A shows the limit element **613** and its corresponding fixer **655**. In this embodiment, the limit element **613** is a gear (hereinafter referred to as a fix gear), and the fixer **655** has a shape corresponding to the teeth profile of the fix gear. Specifically, the fixer **655** may have protruding parts that can be properly filled into the tooth space of the fix gear (for example, the distribution of the protruding parts is the same as the tooth space, and the width of the protruding parts is the same as that of the tooth space). Under this configuration, when the fixer **655** is combined with the limit element **613**, the fix gear can be fixed so that it cannot rotate (refer to FIG. 7B), so as to fix the control gear **611** disposed of in the gear plate **510** to implement the first operation mode of the transmission assembly **190** illustrated in FIG. 8A and FIG. 8B.

(37) It should be noted that although the limit element **613** is shown as gear in this embodiment, the limit element **613** may be in other forms in different embodiments. For example, referring to FIG. 7C, the limit element **613** may be in the form of a pinhole, and the fixer **655** may be in the form of a pin of any form that can achieve the purpose of fixing the limit element **613** by the fixer **655**, which is not limited in the present invention.

(38) FIG. 8A and FIG. 8B illustrate the first operation mode of the transmission assembly **190** of the above-mentioned embodiment (for the second operation mode, please refer to the related descriptions of FIG. 8C and FIG. 8D). In this embodiment, the driving module **650** further includes a driving gear **653** disposed around the gear plate **510**. When the driving module **650** is driven by power, the driving gear **653** will be rotated, and the driving gear **653** can mesh with the gear plate **510**. The foregoing description has explained that when the transmission assembly **190** operates, it will drive the first take-up reel **130** and the first supply reel **150** to rotate, thereby achieving the purpose of replacing the middle part **171** of the first cleaning film **170** covering the outer surface **111**. The operation of the transmission assembly **190** includes rotating the take-up gear **513** and the supply gear **515**, and the structure and process thereof will be described below.

(39) Referring to FIG. 8A and FIG. 8B, since the control gear **611** meshes with the take-up gear **513** and the supply gear **515**, when the driving gear **653** meshes with the gear plate **510** and rotates, and the limiting element **613** of the control element **610** is fixed so that the control gear **611** cannot be rotated, the take-up gear **513** and the supply gear **515** start to rotate relative to the fixed control gear **611** (i.e., revolve around the control gear **611**) in response to the rotation of the gear plate **510**. Due to the meshing relationship between the take-up gear **513** and the supply gear **515** and the control gear **611**, the fixed control gear **611** will drive the take-up gear **513** and the supply gear **515** to rotate (self-rotation), thereby making the transmission assembly **190** operate and perform the take-up operation.

(40) It should be noted that, in this embodiment, the number of teeth of the supply gear **515** can be greater than that of the take-up gear **513**, so that the rotational speed of the take-up gear **513** and the first take-up reel **130** can be faster than that of the supply gear **515** and the first supply reel **150**. Under this configuration, the speed at which the first take-up reel **130** takes up the middle part **171** is faster than the speed at which the first supply reel **150** provides a new middle part **171** so that the first cleaning film **170** can be closely attached to the outer surface **111**.

(41) Referring to FIG. 8C and FIG. 8D, the second operation mode of the transmission assembly **190** of the cleaning equipment **400** according to the embodiment of the present invention is illustrated. In this embodiment, the driving gear **653** of the driving module **650** fixes the gear plate **510** so that it cannot rotate, the driving module **650** drives the control gear **611** to rotate, and the control gear **611** simultaneously meshes with the take-up gear **513** and the supply reel gear **515** to rotate to achieve the purpose of replacing the first cleaning film **170** covering the outer surface **111**. It should be noted that, as described in the third embodiment, the number of teeth of the supply gear **515** may be greater than that of the take-up gear **513**, so that the rotational speed of the take-up

gear **513** and the first take-up reel **130** is faster than that of the supply gear **515** and the first supply reel **150**. In this configuration, the speed at which the first take-up reel **130** takes up the middle part **171** is faster than the speed at which the first supply reel **150** provides a new middle part **171**, so that the first cleaning film **170** can be closely attached to the outer surface **111**.

(42) FIG. **9A** continues illustrating the cleaning equipment **300** according to the third embodiment of the present invention. FIG. **9A** shows the side structure of the cleaning equipment **300**. In this embodiment, the cleaning equipment **300** further includes a lifting module **910** for lifting the frame **110**. Specifically, the lifting module **910** includes a supporting element **911** in the shape of a hollow semi-circle, and the diameter of the semi-circle is slightly larger than that of the gear plate **510** so that the hollow part of the semi-circle can properly accommodate and support the gear plate **510**. When the lifting module **910** is driven by power to rise, the supporting element **911** will exert an upward force on the gear plate **510**, thereby lifting the frame **110**. It should be noted that although the supporting element **911** is shown in the form of a hollow semi-circle in this embodiment, in different embodiments, the supporting element **911** can be in other shapes that can properly support the gear plate **510** (for example, a three-quarters circle), which is not limited in the present invention.

(43) FIG. **9B** continues illustrating the cleaning equipment **300** according to the third embodiment of the present invention, FIG. **9B** shows a schematic diagram of the lifting module **910** when lifting frame **110**. As shown in the figure, when the lifting module **910** lifts frame **110**, the cleaning wheel **310** no longer contacts frame **110**, and the frame **110** is no longer subjected to the friction force generated by the cleaning wheel **310**, and thus stops rotating. In addition, when the lifting module **910** lifts the frame **110**, the limit element **613** of the transmission assembly **190** is coupled to the fixer **655** of the driving module **650**, and the driving gear **653** of the driving module **650** meshes with the gear plate of the transmission assembly **190** (as shown in FIG. **7B**). In this configuration, when the lifting module **910** and the driving module **650** are driven by power, the lifting module **910** lifts the frame **110**, the driving module **650** is coupled to the transmission assembly **190**, and the driving gear **653** starts to rotate, thereby causing the first take-up reel **130** and the first supply reel **150** to rotate to perform the process of replacing the middle part of the cleaning film as shown in FIG. **8A** and FIG. **8B**.

(44) FIG. **10A** to FIG. **10D** illustrate the cleaning equipment **300** according to the third embodiment of the present invention. In this embodiment, the cleaning equipment **300** further includes a monitoring unit **1010**, and the monitoring unit **1010** may include a sensor such as a camera. FIG. **10A** to FIG. **10D** show schematic diagrams of the procedure for replacing the middle part **171** covering the outer surface **111** according to the monitoring result of the monitoring unit **1010**. The sensing range of the monitoring unit **1010** includes at least part of the outer surface **111**, and thus the monitoring unit **1010** can capture an image containing the middle part **171** covering at least part of the outer surface **111**. According to the image, the monitoring unit **1010** can further determine the adsorptive capacity of the middle part **171** of the first cleaning film **170**. When the monitoring unit **1010** determines that the adsorptive capacity of the middle part **171** is smaller than a preset threshold, the lifting module **910** and the driving module **650** will be driven to perform the procedure of replacing the middle part **171** of the first cleaning film **170** as illustrated in FIG. **8A** and FIG. **8B** or FIG. **8C** and FIG. **8D**.

(45) As stated above, when the monitoring unit **1010** determines that the adsorptive capacity of the middle part **171** is smaller than the threshold value (refer to FIG. **10A**), the lifting module **910** lifts frame **110**, and the frame **110** stops rotating in the meantime (refer to FIG. **10B**). Then, the driving module **650** drives the transmission assembly **190** to operate; that is, the driving gear **653** meshes with the gear plate **510** to rotate (and drives the frame **110** to rotate at the same time (refer to FIG. **10C**)), so that the first take-up reel **130** takes up the middle part **171** previously adsorbing the dirt, and the first supply reel **150** provides clean middle part **171** to replace it. Finally, when the middle part of the cleaning film is replaced, the lifting module **910** lowers the frame **110** (refer to FIG.

10D). In this configuration, the monitoring unit **1010** can ensure that the middle part **171** maintains the adsorptive capacity. In addition, since the cleaning wheel **310** continues to rotate during the process of replacing the middle part of the cleaning film, the dirt on the target surface will still be adsorbed by the cleaning wheel **310**. When the lifting module **910** lowers the frame **110**, the middle part contacting the cleaning wheel **310** will adsorb the dirt again to keep the target surface clean. (46) FIG. **11A** to FIG. **11C** illustrate a procedure **1100** for determining the adsorptive capacity of the middle part **171** according to an embodiment of the present invention. At block **1101**, the monitoring unit **1010** captures an image **1120** containing at least part of the middle part **171** (see FIG. **11B**). The monitoring unit **1010** may include an image sensor, such as a complementary metal oxide semiconductor (CMOS) or a charge coupled equipment (CCD) image sensor. Referring to FIG. **11B**, the background of image **1120** is the first cleaning film **170**, and the dots distributed thereon are the dirt adsorbed by the first cleaning film **170**. With continued reference to FIG. **11A**, at block **1103**, appropriate image processing may be performed on image **1120**, such as foreground/background segmentation to reveal the dirt (see processed image **1121** shown in FIG. **11C**). At block **1105**, the condition of the dirt distribution is calculated by object labeling techniques. Then, at block **1107**, the dirt ratio (i.e., the ratio of the dirty area to the area of the first cleaning film **170**) is calculated according to the calculated dirt distribution. At block **1109**, according to the calculated dirt ratio, the monitoring unit **1010** can determine whether the current adsorptive capacity of the middle part **171** of the first cleaning film **170** is smaller than the threshold value (for example, determining that the adsorptive capacity is smaller than the threshold value when the dirt ratio is greater than 85%). At block **1111**, when the monitoring unit **1010** determines that the adsorptive capacity of the middle part **171** of the first cleaning film **170** is smaller than the preset threshold, the procedure of replacing the middle part of the cleaning film as described in the foregoing FIG. **10A** to FIG. **10D** is performed.

(47) Referring to FIG. **12**, the anti-reverse element **1210** according to the third embodiment of the present invention is illustrated. In this embodiment, the take-up gear **513** and the supply gear **515** are rotated counterclockwise (see FIG. **8A** and FIG. **8B**). In order to prevent the take-up gear **513** and the supply gear **515** from rotating in the wrong direction (for example, clockwise), causing the middle part **171** that has adsorbed dirt to cover the outer surface **111** of the frame **110** again, and the clean middle part wind back to the first supply reel **150**, the embodiment further includes the anti-reverse element **1210**. The anti-reverse element **1210** can be arranged beside the take-up gear **513** and/or the supply gear **515**. The anti-reverse element **1210** includes a needle end **1213** and a blocking element **1215**, and the needle end **1213** is arranged in the tooth space. When the take-up gear **513** and/or the supply gear **515** rotate counterclockwise, the needle end **1213** will be moved in the direction opposite to the blocking element **1215** without being blocked by the blocking element **1215**. On the contrary, when the take-up gear **513** and/or the supply gear **515** rotate clockwise, the needle end **1213** will be moved in the direction toward the blocking element **1215**, and thus it will be blocked by the blocking element **1215**, thereby blocking the take-up gear **513** and/or the supply gear **515** rotating clockwise.

(48) It should be noted that, in this embodiment, although the anti-reverse element **1210** is arranged in a manner to block the clockwise rotation of the take-up gear **513** and/or the supply gear **515**, in different embodiments, the anti-reverse element **1210** can be arranged in different ways as required (e.g., in a manner of blocking the counterclockwise rotation of the take-up gear **513** and/or the supply gear **515**), which is not limited in the present invention. In addition, the anti-reverse element **1210** can be arranged beside the gear other than the retake-up gear **513** and/or the supply gear. For example, referring to the embodiment of FIG. **8C** and FIG. **8D**, the anti-reverse element **210** can be arranged beside the control gear **611** to prevent it from rotating counterclockwise.

(49) Referring to FIG. **13**, the fifth embodiment of the present invention provides a cleaning method, which may include the method performed by the cleaning equipment in any of the foregoing embodiments. For example, in this embodiment, the method includes at least the

following steps. At step **1301**, the method includes adsorbing the dirty on the target surface with at least one cleaning wheel **310** and then adsorbing the dirty on the cleaning wheel **310** with the middle part **171**. At step **1303**, the method includes determining, with the monitoring unit **1010**, the state of the middle part **171**, and generating a determination result. At step **1305**, the method includes driving the driving module **650** and the lifting module **910** according to the determination result that the adsorptive capacity of the middle part **171** is less than the threshold value. At step **1307**, the lifting module **910** lifts the frame **110**, while the driving module **650** drives the transmission assembly **190** to replace the middle part **171**. At step **1309**, the lifting module **910** lowers the frame **110** to complete the replacement procedure of the cleaning film.

(50) Although the preferred embodiments of the invention have been described herein, the above description is merely illustrative. The preferred embodiments disclosed will not limit the scope of the invention. Further modification of the invention herein disclosed will occur to those skilled in the respective arts and all such modifications are deemed to be within the scope of the invention as defined by the appended claims.

Claims

1. A cleaning equipment, comprising: a first take-up reel; a first supply reel; a frame comprising an outer surface and an inner space located inside the outer surface, wherein the outer surface has at least one gap communicating with the inner space, and the inner space accommodates the first take-up reel and the first supply reel side-by-side; a first cleaning film with a supply part, a take-up part, and a middle part, wherein the first cleaning film has adsorptive capacity, the supply part is arranged on the first supply reel, the take-up part is arranged on the first take-up reel, at least part of the middle part covers the outer surface of the frame, and two ends of the middle part respectively enter the inner space through the at least one gap and are respectively connected to the supply part and the take-up part; a transmission assembly coupled to the first supply reel and the first take-up reel and configured to drive at least one of the first supply reel and the first take-up reel to rotate; a second take-up reel; a second supply reel; a second cleaning film with a supply part, and a middle art, wherein the second cleaning film has adsorptive capacity, the supply part is arranged on the second supply reel, the take-up part is arranged on the second take-up reel, and at least part of the middle part covers the outer surface of the frame; and wherein the at least one gap includes a first gap and a second gap, the two ends of the middle part of the first cleaning film are respectively connected to the take-up part and the supply part of the first cleaning film through the first gap and the second gap, and two ends of the middle part of the second cleaning film are respectively connected to the take-up part and the supply part of the second cleaning film through the first gap and the second gap.
2. The equipment of claim 1, wherein the outer surface comprises a first outer surface and a second outer surface, wherein the at least part of the middle part of the first cleaning film covers the first outer surface, the at least part of the middle part of the second cleaning film covers the second outer surface, and the first gap and the second gap are respectively located between the first outer surface and the second outer surface.
3. The equipment of claim 1, wherein the transmission assembly includes a belt configured to drive the first supply reel and the first take-up reel to rotate.
4. The equipment of claim 3, further comprising at least one guard plate, wherein the at least one guard plate is coupled to an end of an axial direction of the frame.
5. The equipment of claim 1, wherein the transmission assembly comprises a gear plate, and a surface of the gear plate facing the frame accommodates at least one supply gear and at least one take-up gear, wherein the supply gear and the take-up gear respectively have a first tenon/mortise element corresponding to a second tenon/mortise element of the first supply reel and the first take-up reel, and the number of teeth of the supply gear is greater than that of the take-up gear.

6. The equipment of claim 5, wherein the transmission assembly further comprises a top plate coupled to the gear plate, the top plate has at least two top-plate holes corresponding to the supply gear and the take-up gear, wherein the first tenon/mortise element is coupled to the second tenon/mortise element through the top-plate holes.
7. The equipment of claim 5, wherein the transmission assembly comprises a control element, a first end of the control element is provided with a control gear configured to mesh with the supply gear and the take-up gear respectively, and second end of the control element passes through a gear-plate hole of the gear plate and protrudes from a surface of the gear plate without accommodating the supply gear and the take-up gear.
8. The equipment of claim 7, wherein the second end of the control element is provided with a limit element, and the limit element is a pinhole or a fix gear.
9. The equipment of claim 8, further comprising a driving module and a lifting module, wherein the lifting module has a supporting element corresponding to the gear plate, wherein when the lifting module operates, the supporting element is in contact with the gear plate to lift the frame, and the driving module and the transmission assembly are power coupled.
10. The equipment of claim 9, wherein the driving module comprises a driving gear and a fixer, wherein when the lifting module lifts the frame, the driving gear is adapted to mesh with the gear plate to rotate, and the fixer fixes the limit element of the control element.
11. The equipment of claim 9, wherein the driving module comprises a driving gear, wherein when the lifting module lifts the frame, the driving gear fixes the gear plate, and the driving module drives the control gear to rotate so that the control gear respectively meshes with the supply gear and the take-up gear to rotate.
12. The equipment of claim 1, further comprising at least one cleaning wheel, wherein the cleaning wheel is disposed of under the frame, the axial direction of the cleaning wheel is parallel to the axial direction of the frame, and the cleaning wheel corresponds to the outer surface to contact the at least part of the middle part of the first cleaning film, wherein the adsorptive capacity of the at least one cleaning wheel is smaller than that of the first cleaning film.
13. The equipment of claim 1, further comprising a monitoring unit facing the outer surface of the frame to monitor the state of the at least part of the middle part of the first cleaning film.
14. The equipment of claim 1, further comprising at least one anti-stick strip located at the at least one gap of the frame, wherein the at least one anti-stick strip does not protrude from the outer surface of the frame, and the extending direction of the anti-stick strip is parallel to the axial direction of the frame.
15. The equipment of claim 7, wherein the transmission assembly further comprises an anti-reverse element disposed beside the supply gear, the take-up gear, or the control gear, wherein the anti-reverse element has no effect when the supply gear, the take-up gear or the control gear rotates in a first direction, and the anti-reverse element prevents the supply gear, the take-up gear, or the control gear from rotating when the supply gear, the take-up gear or the control gear rotates in a second direction.
16. A cleaning method, comprising: pulling at least part of a first cleaning film arranged on a first supply reel out to leave an inner space of a frame through at least one gap of the frame, cover an outer surface of the frame, and then enter the inner space again through the at least one gap and be arranged on a first take-up reel, wherein the part of the first cleaning film arranged on the first supply reel becomes a supply part, the part covering the outer surface becomes a middle part, and the part arranged on the first take-up reel becomes a take-up part, the first supply reel and the first take-up reel are arranged side by side in the inner space, and the first cleaning film has absorptive capacity; operating a transmission assembly coupled to the first supply reel and the first take-up reel to drive at least one of the first supply reel and the first take-up reel to rotate, wherein in response to rotation of the first take-up reel, at least part of the middle part becomes the take-up part, and in response to rotation of the first supply reel, at least part of the supply part leaves the

first supply reel and becomes at least part of the middle part to cover the outer surface of the frame; and pulling at least part of a second cleaning film arranged on a second supply reel out to leave the inner space of the frame through the at least one gap of the frame, cover the outer surface of the frame, and then enter the inner space again through the at least one gap and be arranged on a second take-up reel, wherein the part of the second cleaning film arranged on the second supply reel becomes a supply part, the part covering the outer surface becomes a middle part, and the part arranged on the second take-up reel becomes a take-up part, the second supply reel and the second take-up reel are arranged side by side in the inner space, and the second cleaning film has absorptive capacity.

17. The method of claim 16, wherein the at least one gap comprises a first gap and a second gap, wherein the two ends of the middle part of each of the first cleaning film and the second cleaning film are respectively connected to the take-up part and the supplying part of the first cleaning film or the second cleaning film through the first gap and the second gap.

18. The method of claim 17, wherein the outer surface comprises a first outer surface and a second outer surface, wherein the at least part of the middle part of the first cleaning film covers the first outer surface, the at least part of the middle part of the second cleaning film covers the second outer surface, and the first gap and the second gap are respectively located between the first outer surface and the second outer surface.

19. The method of claim 16, wherein the transmission assembly comprises a belt, and when at least one of the first supply reel and the first take-up reel is driven to rotate, the belt is configured to drive the not driven one of the first supply reel and the first take-up reel to rotate.

20. The method of claim 19, further comprising coupling at least one guard plate to an end of an axial direction of the frame.

21. The method of claim 16, wherein operating the transmission assembly to drive at least one of the first supply reel and the first take-up reel to rotate comprises: using a control gear disposed at first end of a control element of the transmission assembly to mesh with a supply gear of the first supply reel, and using the control gear to mesh with a take-up gear coupled to the first take-up reel; and driving the control gear to rotate by a driving module, so as to drive the first supply reel and the first take-up reel to rotate.

22. The method of claim 16, wherein operating the transmission assembly to drive at least one of the first supply reel and the first take-up reel to rotate comprises: accommodating a supply gear and a take-up gear by a gear plate of the transmission assembly, wherein the supply gear is coupled to the first supply reel and the take-up gear is coupled to the first take-up reel; using a control gear disposed at first end of a control element of the transmission assembly to mesh with the supply gear and the take-up gear; fixing the control element and driving the gear plate to rotate by a driving module, wherein the supply gear and the take-up gear rotate relative to the fixed control gear, and in response to the meshing relationship between them, the supply gear and the take-up gear rotates to drive the first supply reel and the first take-up reel to rotate, wherein the rotational speed of the first take-up reel is greater than that of the first supply reel.

23. The method of claim 21, wherein driving the first supply reel and the first take-up reel to rotate by the driving module further comprises: monitoring the state of the middle part of the first cleaning film by a monitoring unit to determine whether the adsorptive capacity of the middle part is less than a threshold; lifting the frame by the lifting module in response to the determination that the adsorptive capacity of the middle part is less than the threshold so that the driving module and the transmission assembly are power coupled.

24. The method of claim 23, further comprising driving a cleaning wheel to rotate with power, wherein when the frame is not lifted, at least part of the cleaning wheel contacts the at least part of the middle part of the first cleaning film, to drive the frame to rotate.

25. The method of claim 24, further comprising adsorbing dirt with the cleaning wheel, wherein the adsorptive capacity of the cleaning wheel is smaller than that of the first cleaning film, and the

cleaning wheel passes the adsorbed dirt on to the first cleaning film through the at least part of the middle part in contact with the cleaning wheel.

26. The method of claim 25, wherein when the frame is lifted, the cleaning wheel does not contact the first cleaning film, and the cleaning wheel continues to rotate and absorb dirt.

27. The method of claim 16, further comprising disposing at least one anti-stick strip in the at least one gap, wherein the at least one anti-stick strip does not protrude from the outer surface of the frame, and the extending direction of the anti-stick strip is parallel to the axial direction of the frame.

28. The method of claim 21, further comprising disposing an anti-reverse element beside the supply gear, the take-up gear, or the control gear, wherein the anti-reverse element has no effect when the supply gear, the take-up gear or the control gear rotates in a first direction, and the anti-reverse element prevents the supply gear, the take-up gear, or the control gear from rotating when the supply gear, the take-up gear or the control gear rotates in a second direction.

29. A cleaning equipment, comprising: a first take-up reel; a first supply reel; a frame comprising an outer surface and an inner space located inside the outer surface, wherein the outer surface has at least one gap communicating with the inner space, and the inner space accommodates the first take-up reel and the first supply reel side-by-side; a first cleaning film with a supply part, a take-up part, and a middle part, wherein the first cleaning film has adsorptive capacity, the supply part is arranged on the first supply reel, the take-up part is arranged on the first take-up reel, at least part of the middle part covers the outer surface of the frame, and two ends of the middle part respectively enter the inner space through the at least one gap and are respectively connected to the supply part and the take-up part; a transmission assembly coupled to the first supply reel and the first take-up reel and configured to drive at least one of the first supply reel and the first take-up reel to rotate; and at least one cleaning wheel, wherein the cleaning wheel is disposed of under the frame, the axial direction of the cleaning wheel is parallel to the axial direction of the frame, and the cleaning wheel corresponds to the outer surface to contact the at least part of the middle part of the first cleaning film, wherein the adsorptive capacity of the at least one cleaning wheel is smaller than that of the first cleaning film.
